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RESEARCH INTERESTS

Memristor device engineering for compute-in-memory and neuromorphic functions (synapse/logic-in-memory), focusing on low-power switching, variability, and reliability.

DRAM MIM capacitor development (high-k dielectrics/electrodes/stack integration) with emphasis on scaling, leakage control, and reliability qualification.

Thin-film process integration and electrical characterization (ALD-based stacks; I - V / C - V /pulse testing) with interface optimization.

Physics-guided analysis and modeling of charge transport and trap/defect dynamics in thin films to improve performance and stability.

EDUCATION

2023.03 – 2028.02 (Expected)

Korea University, School of Electrical Engineering

Integrated M.S.–Ph.D. Program

Advisor: Jae Won Shim

PUBLICATIONS

[First-author publications]

1. **Min Jong Lee**[†], Hyoungwook Cho[†], Sang Heon Lee, Ohhyun Kwon, Minil Kang, Dong Gyu Lee, Seongwon Lee, Tae Kyung Lee, Gyeong Min Lee, Minseong Um, Sung Su Yoon, Gayoung Ham, Hyojung Cha, Hyungju Ahn, SungWoo Nam, Hyung-Min Lee*, and Jae Won Shim*, “Helicity-defined analog conductance states in chiral perovskite synapses” *Advanced Materials* (2026) (Accepted)
2. **Min Jong Lee**, Sang Heon Lee, Dong Gyu Lee, Tae Hyuk Kim, Yubhin Cho, Gyeong Min Lee, Sung Su Yoon, Seon Joong Kim, Hyungju Ahn, Tae Kyung Lee*, and Jae Won Shim*, “Linear and Symmetric Artificial Synapses Driven by Hydrogen Bonding for Accurate and Reliable Neuromorphic Computing” *Advanced Materials* 37, 45, e11728 (2025)
3. **Min Jong Lee**, Tae Hyuk Kim, Sang Heon Lee, Seunghyun Oh, Muhammad Asghar Khan, Gyeong Min Lee, Young Kyun Choi, Soyeon Lee, Hyungju Ahn, Soong Ju Oh, Jiwoong Yang, and Jae Won Shim*, “Intermediate Layer-Assisted Trap Density Reduction in Low-Power Optoelectronic Memristors for Multifunctional Systems” *Advanced Functional Materials* 35, 22, 2421080 (2025)
4. **Min Jong Lee**, Seunghyun Oh, Muhammad Ahsan Saeed, Yelim Kang, Gyeong Min Lee, Hyungju Ahn, Yunsang Kim, and Jae Won Shim*, “Fine-tuning Hole Collection via Metal Pseudo-halogenated Thin-film Treatment for Propelling Advancements in Versatile platforms” *IEEE Transactions on Electron Devices* 71, 8, 4723-4731 (2024)
5. **Min Jong Lee**, Ji-Sang Park, Tae Hyuk Kim, Muhammad Ahsan Saeed, Gyeong Min Lee, and Jae Won Shim*, “Tailoring Hole-Selective Contacts via Self-Assembled Monolayers for advancing Indoor Organic Photovoltaic and Capacitor Devices” *Chemical Engineering Journal* 481, 148481 (2024)
6. **Min Jong Lee**, Muhammad Ahsan Saeed, Tae Hyuk Kim, Jae Jin Chung, and Jae Won Shim*, “Atomic layer deposition-processed laminated In_2O_3 : V_2O_5 nanocompounds: Versatile electrode platforms for high-performance organic photovoltaics and capacitor device” *Journal of Alloys and Compounds* 968, 171951 (2023)

[Co-author publications]

7. Sang Heon Lee[†], Seung Su Shin[†], Yubhin Cho[†], **Min Jong Lee**, Sung Su Yoon, Dong Gyu Lee, Seon Joong Kim, Seunghyun Oh, Ohhyun Kwon, Gyeong Min Lee, Hyungju Ahn, Tae Kyung Lee*, Jung-Hyun Lee* and Jae Won Shim*, “Harnessing Selective Defect Formation for Filament Control in Reliable and Energy-Efficient Perovskite Resistive Random Access Memory” *Advanced Functional Materials* (2026) (Early View)

8. Seon Joong Kim[†], Junhyeok Park[†], Byung Ku Jung, Ohhyun Kwon, Sang Heon Lee, Hyung Jin Choi, Seongyong Hong, **Min Jong Lee**, Soong Ju Oh*, and Jae Won Shim*, “Defect-Healed Perovskite Quantum Dots for Optoelectronic Applications in Indoor Photovoltaics and Photoplethysmography” *Small* 22, 50, e74660 (2026)
9. Sung Su Yoon, Sang Heon Lee, **Min Jong Lee**, Seon Joong Kim, Hyungju Ahn, Han Jung Park and Jae Won Shim*, “Ionic Landscape Engineering via Perovskite Quantum Dots for Reliable and Energy-Efficient Perovskite Memristors” *ACS Nano* 20, 21, 15336-15346 (2026)
10. Ohhyun Kwon[†], Jae Hoon Son[†], Tae Hyuk Kim, Gyeong Min Lee, Seunghyun Oh, Gayoung Ham, **Min Jong Lee**, Sang Heon Lee, Seon Joong Kim, Hyojung Cha, Han Young Woo*, and Jae Won Shim*, “Suppression of Photo-Mediated Traps in Integrated Organic Photovoltaic–Photodetector Devices via N-Type Self-Assembly-Driven Interfacial Engineering” *Advanced Functional Materials* 36, 28, e24759 (2026)
11. Minju Yang, Seon Joong Kim, Tae Hyuk Kim, Hyungju Ahn, **Min Jong Lee**, Yunsang Kim, and Jae Won Shim*, “Bi-Synergistic Ligand-Mediated Passivation of Surface Defects for Highly Efficient and Stable Cesium-Lead-Iodide Perovskite Quantum Dot Photovoltaics” *Materials Horizons* 13, 1906 - 1917 (2026)
12. Hee Chun Kim,[†] Ji Hyeon Lee,[†] Tae Hyuk Kim, **Min Jong Lee**, Eun Soo Shim, Seunghwan Bae, Hyungju Ahn, Han Jung Park, Nak-Seung Patrick Hyun, Jea Woong Jo*, and Jae Won Shim*, “Molecular Spacer Engineering of Carbazole–Phosphonic Acid SAMs: Probing Interfacial Energetics in Low-Light Organic Optoelectronic Devices” *Chemical Engineering Journal* 527, 171902 (2026)
13. Muhammad Asghar Khan, Yelim Kang, Muhammad Farooq Khan, Shania Rehman, **Min Jong Lee**, Sang Heon Lee, Seunghyun Oh, Tae Hyuk Kim, Seon Joong Kim, Hyungju Ahn, and Jae Won Shim*, “Ultraviolet light-Driven Artificial Neuromorphic Properties in Organic Transistors for Reservoir Computing and Encrypted Communication” *IEEE Transactions on Electron Devices* 72, 11, 6341-6348 (2025)
14. Sang Heon Lee, **Min Jong Lee**, Hyungju Ahn, Ji-Sang Park, Seon Joong Kim, Kihyun Lee, Tae Hyuk Kim, Gyeong Min Lee, Kwanpyo Kim, and Jae Won Shim*, “Synaptic Plasticity Modulation in Dion–Jacobson Perovskite Artificial Synapses Enabled by Crystallographic Orientation Control” *Small* 21, 34, 2504328 (2025)
15. Yelim Kang, Tae Hyuk Kim, **Min Jong Lee**, and Jae Won Shim*, “Dual-Mode Organic Photodetectors with Optimized Charge Dynamics and Barrier Modulation for Superior Spectral Selectivity” *Small* 21, 29, 2503701 (2025)
16. Tae Hyuk Kim,[†] Sang Young Jeong,[†] Seunghyun Oh, Yelim Kang, **Min Jong Lee**, Min Hun Jee, Han Young Woo*, and Jae Won Shim*, “Control of chemical doping-mediated space charge for energetic band-switching modulation in low-noise shortwave-infrared organic photodetector” *Advanced Materials* 37, 30, 2500126 (2025)
17. Seon Joong Kim, Muhammad Asghar Khan, Dong Gyu Lee, Tae Hyuk Kim, **Min Jong Lee**, Yelim Kang, Minju Yang, Hyungju Ahn, Kyoungwon Park, Tae Kyung Lee*, and Jae Won Shim*, “Strategic Selection of Functional Groups for Ligand Passivation to Improve Performance of Cesium–Lead–Iodide Quantum Dot Photovoltaics and Luminescent Solar Concentrators” *Advanced Functional Materials* 35, 33, 2423796 (2025)
18. Gyeong Min Lee, Tae Hyuk Kim, Gayoung Ham, **Min Jong Lee**, Seunghyun Oh, Yelim Kang, Hyungju Ahn, Hyojung Cha, Seungjun Chung, and Jae Won Shim*, “Enhancing Dynamic Range in Low-Noise 2D-Integrated Organic Photodiodes by Mitigating Langevin Recombination” *ACS Nano* 19, 4, 4650–4662 (2025)
19. Muhammad Ahsan Saeed[†], Zubair Khalid[†], **Min Jong Lee**, Hyungju Ahn, Jae-Min Oh*, and Jae Won Shim*, “Bidentate mode coordinated Ti3CNTx MXene-controlled interfacial engineering enables high-performance indoor organic photovoltaic devices” *Chemical Engineering Journal* 495, 153404 (2024)
20. Tae Hyuk Kim[†], Ji Hyeon Lee[†], Min Ho Jang[†], Gyeong Min Lee, Eun Soo Shim, Seunghyun Oh, Muhammad Ahsan Saeed, **Min Jong Lee**, Byoung-Soo Yu, Do Kyung Hwang, Chae Won Park, Sae Youn Lee*, Jea Woong Jo*, and Jae Won Shim*, “Atto-scale noise near-infrared organic photodetectors enabled by controlling interfacial energetic offset through enhanced anchoring ability” *Advanced Materials* 36, 40, 2403647 (2024)
21. Seunghyun Oh[†], Ye Lim Kang[†], Tae Hyuk Kim, Seon Joong Kim, **Min Jong Lee**, Gyeong Min Lee, Muhammad Ahsan Saeed*, and Jae Won Shim*, “Enhancing the indoor performance of organic photovoltaic devices: Interface engineering with an aminobenzoic-acid-based self-assembled monolayer” *Journal of Physics: Energy*, 6, 025015 (2024)
22. Tae Hyuk Kim[†], Young Kyun Choi[†], Gyeong Min Lee, Muhammad Ahsan Saeed, Byung Ku Jung, **Min Jong Lee**, Hyung Jin Choi, Soong Ju Oh*, and Jae Won Shim*, “Ultra-low Noise Level Infrared Quantum Dot Photodiodes with Self-Screenable Polymeric Optical Window” *Advanced Materials* 36,4, 2309028 (2024)

PATENTS

1. 심재원, **이민중**, 최학영, 장성환, “금속 유사할로겐화 박막을 이용한 무기 발광 디스플레이용 이온 차폐막 및 그 제조방법” 등록번호 10-2995909, 등록일자 2026.07.22
2. 심재원, **이민중**, “다기능 저전력 광전자 메모리스트 장치 및 이의 제조 방법” 등록번호 10-2968668, 등록일자 2026.05.22
3. 심재원, **이민중**, “MULTIFUNCTIONAL LOW-POWER OPTOELECTRONIC MEMRISTORS DEVICE AND METHOD OF MANUFACTURING THEREOF” 출원번호 (US) 19/549,394, 출원일자 2026.02.25

4. 김태균, 심재원, 이민중, “Semiconductor device”, 출원번호 (US) 19/002282, 출원일자 (미국, 중국, 대만) 2025.11.13
5. 김태균, 심재원, 이민중, “METHOD OF MANUFACTURING SEMICONDUCTOR MEMORY DEVICE”, 출원번호 (US) 19/037,592, 출원일자 (미국, 대만) 2025.01.27
6. 심재원, 이민중, “페로브스카이트-고분자 박막을 이용한 선형 대칭형 인공 시냅스 소자 및 이의 제조 방법” 출원번호 10-2025-0091536, 출원일자 2025.07.08
7. 김태균, 심재원, 이민중, “반도체 메모리 소자의 제조 방법”, 출원번호 10-2024-0088513, 출원일자 2024.07.04
8. 김태균, 심재원, 이민중, “반도체 장치”, 출원번호 10-2024-0060762, 출원일자 2024.05.08

HONORS/AWARDS

- 고려대 공과대학 연구실적 우수대학원생 수상 (2026)
- 삼성전자 산학협력 우수논문 우수상 (2025)
- 한국이공학진흥원장 차세대 공학연구자 우수상 (2025)
- BK21 참여대학원생 우수저널 게재 논문 선정 (2026)
- KU 희망장학금 선정 (2026)
- 한국이공학진흥원장 제 2기 IPESK 차세대 공학 연구자 선정 (2025)

PROJECTS

1. 박사과정생 연구장려금지원사업(연구책임자), “상대 형성 동역학 및 저주파 잡음 해석 기반 이온 이동 제어형 메모리스터의 멀티 레벨 정밀도 한계 규명”, 교육부, (2026.09-현재)
2. 기초연구실, “초경량 비행형 Physical AI 를 위한 Photonic Skin 연구실”, 과학기술정보통신부, (2026.07-현재)
3. 삼성전자, “초고굴곡 FPCB 배선 소재 개발”, 삼성전자(주), (2026.01-현재)
4. 시장선도를 위한 한국 주도형 K-Sensor 기술개발 (R&D), “스마트 주거 환경을 위한 저전력 다중 복합 유해가스 측정용 적외선 가스센서 기술 개발”, 산업통상자원부, (2024.04-현재)
5. 중견 연구, “전일 동작 광전지 개발을 위한 약광 발전 한계 돌파 연구”, 과학기술정보통신부, (2022.03-현재)
6. 삼성전자 전략산학, “TiN 전극 대체를 위한 atomic layer deposition (ALD) 기반 초박막 다성분계 (다층박막) 전극 개발”, 삼성전자(주), (2020.09-2025.09)
7. 산업혁신인재성장지원(R&D), “산업 전환형 무기발광 디스플레이 전문인력양성사업”, 산업통상자원부, (2023.03-2025.08)
8. 전략 핵심 소재 자립화 기술개발, “고신축특성을 가지는 고신뢰성 고내구성 점착필름 및 대면적 코팅 공정 기술 개발”, 산업통상자원부, (2020.04-2024.12)
9. 삼성전자, “Liquid Metal Particle 활용 Stretchable PCB 제작”, 삼성전자(주), (2024.08-2024.11)

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